Oxidative Phosphorylation and Photophosphorylation

chapter **19**

1. Oxidation-Reduction Reactions The NADH dehydrogenase complex of the mitochondrial respiratory chain promotes the following series of oxidation-reduction reactions, in which Fe^{3+} and Fe^{2+} represent the iron in iron-sulfur centers, Q is ubiquinone, QH_2 is ubiquinol, and E is the enzyme:

(1) $NADH + H^+ + \text{E-FMN} \longrightarrow NAD^+ + \text{E-FMNH}_2$

(2) $\text{E-FMNH}_2 + 2Fe^{3+} \longrightarrow \text{E-FMN} + 2Fe^{2+} + 2H^+$

(3) $2Fe^{2+} + 2H^+ + Q \longrightarrow 2Fe^{3+} + QH_2$

Sum: $NADH + H^+ + Q \longrightarrow NAD^+ + QH_2$

For each of the three reactions catalyzed by the NADH dehydrogenase complex, identify **(a)** the electron donor, **(b)** the electron acceptor, **(c)** the conjugate redox pair, **(d)** the reducing agent, and **(e)** the oxidizing agent.

Answer Oxidation-reduction reactions require an electron donor and an electron acceptor. Recall that electron donors are reducing agents; electron acceptors are oxidizing agents.

(1) NADH is the electron donor **(a)** and the reducing agent **(d);** E-FMN is the electron acceptor **(b)** and the oxidizing agent **(e);** NAD^+/NADH and E-FMN/E-FMNH_2 are conjugate redox pairs **(c).**

(2) E-FMNH_2 is the electron donor **(a)** and reducing agent **(d);** Fe^{3+} is the electron acceptor **(b)** and oxidizing agent **(e);** E-FMN/E-FMNH_2 and Fe^{3+}/Fe^{2+} are redox pairs **(c).**

(3) Fe^{2+} is the electron donor **(a)** and reducing agent **(d);** Q is the electron acceptor **(b)** and oxidizing agent **(e);** and Fe^{3+}/Fe^{2+} and Q/QH_2 are redox pairs **(c).**

2. All Parts of Ubiquinone Have a Function In electron transfer, only the quinone portion of ubiquinone undergoes oxidation-reduction; the isoprenoid side chain remains unchanged. What is the function of this chain?

Answer The long isoprenoid side chain makes ubiquinone very soluble in lipids and allows it to diffuse in the semifluid membrane. This is important because ubiquinone transfers electrons from Complexes I and II to Complex III, all of which are embedded in the inner mitochondrial membrane.

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3. Use of FAD Rather Than NAD^+ in Succinate Oxidation All the dehydrogenases of glycolysis and the citric acid cycle use NAD^+ ($E'°$ for NAD^+/NADH is -0.32 V) as electron acceptor except succinate dehydrogenase, which uses covalently bound FAD ($E'°$ for FAD/$FADH_2$ in this enzyme is 0.050 V). Suggest why FAD is a more appropriate electron acceptor than NAD^+ in the dehydrogenation of succinate, based on the $E'°$ values of fumarate/succinate ($E'° = 0.031$), NAD^+/NADH, and the succinate dehydrogenase FAD/$FADH_2$.

Answer From the difference in standard reduction potential ($\Delta E'°$) for each pair of half-reactions, we can calculate the $\Delta G'°$ values for the oxidation of succinate using NAD^+ and oxidation using E-FAD.

For NAD^+:

$$\begin{aligned}\Delta G'° &= -n\mathcal{F}\Delta E'° \\ &= -2(96.5\ \text{kJ/V} \cdot \text{mol})(-0.32\ \text{V} - 0.031\ \text{V}) \\ &= \ \ 68\ \text{kJ/mol}\end{aligned}$$

For E-FAD:

$$\begin{aligned}\Delta G'° &= -2(96.5\ \text{kJ/V} \cdot \text{mol})(0.050\ \text{V} - 0.031\ \text{V}) \\ &= -3.7\ \text{kJ/mol}\end{aligned}$$

The oxidation of succinate by E-FAD is favored by the negative standard free-energy change, which is consistent with a K'_{eq} of >1. Oxidation by NAD^+ would require a large, positive, standard free-energy change and have a K'_{eq} favoring the synthesis of succinate.

4. Degree of Reduction of Electron Carriers in the Respiratory Chain The degree of reduction of each carrier in the respiratory chain is determined by conditions in the mitochondrion. For example, when NADH and O_2 are abundant, the steady-state degree of reduction of the carriers decreases as electrons pass from the substrate to O_2. When electron transfer is blocked, the carriers before the block become more reduced and those beyond the block become more oxidized (see Fig. 19–6). For each of the conditions below, predict the state of oxidation of ubiquinone and cytochromes *b*, c_1, *c*, and $a + a_3$.

(a) Abundant NADH and O_2, but cyanide added

(b) Abundant NADH, but O_2 exhausted

(c) Abundant O_2, but NADH exhausted

(d) Abundant NADH and O_2

Answer As shown in Figure 19–6, the oxidation-reduction state of the carriers in the electron-transfer system varies with the conditions.

(a) Cyanide inhibits cytochrome oxidase ($a + a_3$); all carriers become reduced.

(b) In the absence of O_2, no terminal electron acceptor is present; all carriers become reduced.

(c) In the absence of NADH, no carrier can be reduced; all carriers become oxidized.

(d) These are the usual conditions for an aerobic, actively metabolizing cell; the early carriers (e.g., Q) are somewhat reduced, while the late ones (e.g., cytochrome *c*) are oxidized.

5. Effect of Rotenone and Antimycin A on Electron Transfer Rotenone, a toxic natural product from plants, strongly inhibits NADH dehydrogenase of insect and fish mitochondria. Antimycin A, a toxic antibiotic, strongly inhibits the oxidation of ubiquinol.

(a) Explain why rotenone ingestion is lethal to some insect and fish species.

(b) Explain why antimycin A is a poison.

(c) Given that rotenone and antimycin A are equally effective in blocking their respective sites in the electron-transfer chain, which would be a more potent poison? Explain.

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Answer

(a) The inhibition of NADH dehydrogenase by rotenone decreases the rate of electron flow through the respiratory chain, which in turn decreases the rate of ATP production. If this reduced rate is unable to meet its ATP requirements, the organism dies.

(b) Antimycin A strongly inhibits the oxidation of reduced Q in the respiratory chain, severely limiting the rate of electron transfer and ATP production.

(c) Electrons flow into the system at Complex I from the NAD^+-linked reactions and at Complex II from succinate and fatty acyl–CoA through FAD (see Figs. 19–8 and 19–16). Antimycin A inhibits electron flow (through Q) from *all* these sources, whereas rotenone inhibits flow only through Complex I. Thus, antimycin A is a more potent poison.

6. Uncouplers of Oxidative Phosphorylation In normal mitochondria the rate of electron transfer is tightly coupled to the demand for ATP. When the rate of use of ATP is relatively low, the rate of electron transfer is low; when demand for ATP increases, electron-transfer rate increases. Under these conditions of tight coupling, the number of ATP molecules produced per atom of oxygen consumed when NADH is the electron donor—the P/O ratio–is about 2.5.

(a) Predict the effect of a relatively low and a relatively high concentration of uncoupling agent on the rate of electron transfer and the P/O ratio.

(b) Ingestion of uncouplers causes profuse sweating and an increase in body temperature. Explain this phenomenon in molecular terms. What happens to the P/O ratio in the presence of uncouplers?

(c) The uncoupler 2,4-dinitrophenol was once prescribed as a weight-reducing drug. How could this agent, in principle, serve as a weight-reducing aid? Uncoupling agents are no longer prescribed because some deaths occurred following their use. Why might the ingestion of uncouplers lead to death?

Answer Uncouplers of oxidative phosphorylation stimulate the rate of electron flow but not ATP synthesis.

(a) At relatively low levels of an uncoupling agent, P/O ratios drop somewhat, but the cell can compensate for this by increasing the rate of electron flow; ATP levels can be kept relatively normal. At high levels of uncoupler, P/O ratios approach zero and the cell cannot maintain ATP levels.

(b) As amounts of an uncoupler increase, the P/O ratio decreases and the body struggles to make sufficient ATP by oxidizing more fuel. The heat produced by this increased rate of oxidation raises the body temperature. The P/O ratio is affected as noted in **(a).**

(c) Increased activity of the respiratory chain in the presence of an uncoupler requires the degradation of additional energy stores (glycogen and fat). By oxidizing more fuel in an attempt to produce the same amount of ATP, the organism loses weight. If the P/O ratio nears zero, the lack of ATP will be lethal.

7. Effects of Valinomycin on Oxidative Phosphorylation When the antibiotic valinomycin is added to actively respiring mitochondria, several things happen: the yield of ATP decreases, the rate of O_2 consumption increases, heat is released, and the pH gradient across the inner mitochondrial membrane increases. Does valinomycin act as an uncoupler or an inhibitor of oxidative phosphorylation? Explain the experimental observations in terms of the antibiotic's ability to transfer K^+ ions across the inner mitochondrial membrane.

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Answer The observed effects are consistent with the action of an uncoupler—that is, an agent that causes the free energy released in electron transfer to appear as heat rather than in ATP. In respiring mitochondria, H^+ ions are translocated out of the matrix during electron transfer, creating a proton gradient and an electrical potential across the membrane. A significant portion of the free energy used to synthesize ATP originates from this electric potential. Valinomycin combines with K^+ ions to form a complex that passes through the inner mitochondrial membrane. So, as a proton is translocated out by electron transfer, a K^+ ion moves in, and the potential across the membrane is lost. This reduces the yield of ATP per mole of protons flowing through ATP synthase (F_oF_1). In other words, electron transfer and phosphorylation become uncoupled. In response to the decreased efficiency of ATP synthesis, the rate of electron transfer increases markedly. This results in an increase in the H^+ gradient, in oxygen consumption, and in the amount of heat released.

8. **Mode of Action of Dicyclohexylcarbodiimide (DCCD)** When DCCD is added to a suspension of tightly coupled, actively respiring mitochondria, the rate of electron transfer (measured by O_2 consumption) and the rate of ATP production dramatically decrease. If a solution of 2,4-dinitrophenol is now added to the preparation, O_2 consumption returns to normal but ATP production remains inhibited.
   (a) What process in electron transfer or oxidative phosphorylation is affected by DCCD?
   (b) Why does DCCD affect the O_2 consumption of mitochondria? Explain the effect of 2,4-dinitrophenol on the inhibited mitochondrial preparation.
   (c) Which of the following inhibitors does DCCD most resemble in its action: antimycin A, rotenone, or oligomycin?

   Answer
   (a) DCCD inhibits ATP synthesis. In tightly coupled mitochondria, this inhibition leads to inhibition of electron transfer also.
   (b) A decrease in electron transfer causes a decrease in O_2 consumption. 2,4-Dinitrophenol uncouples electron transfer from ATP synthesis, allowing respiration to increase. No ATP is synthesized and the P/O ratio decreases.
   (c) DCCD and oligomycin inhibit ATP synthesis (see Table 19–4).

9. **Compartmentalization of Citric Acid Cycle Components** Isocitrate dehydrogenase is found only in the mitochondrion, but malate dehydrogenase is found in both the cytosol and mitochondrion. What is the role of cytosolic malate dehydrogenase?

   Answer Malate dehydrogenase catalyzes the conversion of malate to oxaloacetate in the citric acid cycle, which takes place in the mitochondrion, and also plays a key role in the transport of reducing equivalents across the inner mitochondrial membrane via the malate-aspartate shuttle (Fig. 19–29). This shuttle requires the presence of malate dehydrogenase in the cytosol and the mitochondrial matrix.

10. **The Malate–α-Ketoglutarate Transport System** The transport system that conveys malate and α-ketoglutarate across the inner mitochondrial membrane (see Fig. 19–29) is inhibited by *n*-butylmalonate. Suppose *n*-butylmalonate is added to an aerobic suspension of kidney cells using glucose exclusively as fuel. Predict the effect of this inhibitor on **(a)** glycolysis, **(b)** oxygen consumption, **(c)** lactate formation, and **(d)** ATP synthesis.

    Answer NADH produced in the cytosol cannot cross the inner mitochondrial membrane, but must be oxidized if glycolysis is to continue. Reducing equivalents from NADH enter the mitochondrion by way of the malate-aspartate shuttle. NADH reduces oxaloacetate to form malate and NAD^+, and the malate is transported into the mitochondrion. Cytosolic oxidation of glucose can continue, and the malate is converted back to oxaloacetate and NADH in the mitochondrion (see Fig. 19–29).

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(a) If *n*-butylmalonate, an inhibitor of the malate–α-ketoglutarate transporter, is added to cells, NADH accumulates in the cytosol. This forces glycolysis to operate anaerobically, with reoxidation of NADH in the lactate dehydrogenase reaction.

(b) Because reducing equivalents from the oxidation reactions of glycolysis do not enter the mitochondrion, oxygen consumption slows and eventually ceases.

(c) The end product of anaerobic glycolysis, lactate, accumulates.

(d) ATP is not formed aerobically because the cells have converted to anaerobic glycolysis. Overall, ATP synthesis decreases drastically, to 2 ATP per glucose molecule.

11. Cellular ADP Concentration Controls ATP Formation Although ADP and P_i are required for the synthesis of ATP, the rate of synthesis depends mainly on the concentration of ADP, not P_i. Why?

Answer The steady-state concentration of P_i in the cell is much higher than that of ADP. As the ADP concentration rises as a result of ATP consumption, there is little change in $[P_i]$, so P_i cannot serve as a regulator.

12. Time Scales of Regulatory Events in Mitochondria Compare the likely time scales for the adjustments in respiratory rate caused by **(a)** increased [ADP] and **(b)** reduced pO_2. What accounts for the difference?

Answer In **(a),** respiratory control by ADP, the increase in respiratory rate is limited by the rate of diffusion of ADP, and the response would be expected to occur in fractions of a millisecond. The adjustment to **(b),** hypoxia mediated by HIF-1, requires a change in concentration of several proteins, the result of increased synthesis or degradation. The time scale for protein synthesis or degradation is typically many seconds to hours—much longer than the time required for changes in substrate concentration.

13. The Pasteur Effect When O_2 is added to an anaerobic suspension of cells consuming glucose at a high rate, the rate of glucose consumption declines greatly as the O_2 is used up, and accumulation of lactate ceases. This effect, first observed by Louis Pasteur in the 1860s, is characteristic of most cells capable of aerobic and anaerobic glucose catabolism.

(a) Why does the accumulation of lactate cease after O_2 is added?

(b) Why does the presence of O_2 decrease the rate of glucose consumption?

(c) How does the onset of O_2 consumption slow down the rate of glucose consumption? Explain in terms of specific enzymes.

Answer The addition of oxygen to an anaerobic suspension allows cells to convert from fermentation to oxidative phosphorylation as a mechanism for reoxidizing NADH and making ATP. Because ATP synthesis is much more efficient under aerobic conditions, the amount of glucose needed will decrease (the Pasteur effect). This decreased utilization of glucose in the presence of oxygen can be demonstrated in any tissue that is capable of aerobic and anaerobic glycolysis.

(a) Oxygen allows the tissue to convert from lactic acid fermentation to respiratory electron transfer and oxidative phosphorylation as the mechanism for NADH oxidation.

(b) Cells produce much more ATP per glucose molecule oxidized aerobically, so less glucose is needed.

(c) As [ATP] rises, phosphofructokinase-1 is inhibited, thus slowing the rate of glucose entry into the glycolytic pathway.

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14. Respiration-Deficient Yeast Mutants and Ethanol Production Respiration-deficient yeast mutants (p^-; "petites") can be produced from wild-type parents by treatment with mutagenic agents. The mutants lack cytochrome oxidase, a deficit that markedly affects their metabolic behavior. One striking effect is that fermentation is not suppressed by O_2—that is, the mutants lack the Pasteur effect (see Problem 13). Some companies are very interested in using these mutants to ferment wood chips to ethanol for energy use. Explain the advantages of using these mutants rather than wild-type yeast for large-scale ethanol production. Why does the absence of cytochrome oxidase eliminate the Pasteur effect?

Answer The absence of cytochrome oxidase prevents these mutants from oxidizing the products of fermentation (ethanol, acetate, lactate, or glycerol) via the normal respiratory route. These mutants do not have a working citric acid cycle because they cannot reoxidize NADH through the O_2-dependent electron-transfer chain. Thus, catabolism of glucose stops at the ethanol stage, even in the presence of oxygen. The ability to carry out these fermentations in the presence of oxygen is a major practical advantage because completely anaerobic conditions are difficult to maintain. The Pasteur effect—the decrease in glucose consumption that occurs when oxygen is introduced—is not observed in the absence of an active citric acid cycle and electron-transfer chain.

15. Advantages of Supercomplexes for Electron Transfer There is growing evidence that mitochondrial Complexes I, II, III, and IV are part of a larger supercomplex. What might be the advantage of having all four complexes within a supercomplex?

Answer When electron-carrying complexes are bound together in a supercomplex, electron flow between complexes occurs in a solid state; this electron movement is kinetically favored compared with the situation in which electron flow depends on each complex diffusing to and colliding with the next complex in the chain.

16. How Many Protons in a Mitochondrion? Electron transfer translocates protons from the mitochondrial matrix to the external medium, establishing a pH gradient across the inner membrane (outside more acidic than inside). The tendency of protons to diffuse back into the matrix is the driving force for ATP synthesis by ATP synthase. During oxidative phosphorylation by a suspension of mitochondria in a medium of pH 7.4, the pH of the matrix has been measured as 7.7.

(a) Calculate $[H^+]$ in the external medium and in the matrix under these conditions.

(b) What is the outside-to-inside ratio of $[H^+]$? Comment on the energy inherent in this concentration difference. (Hint: see Eqn 11–4, p. 396)

(c) Calculate the number of protons in a respiring liver mitochondrion, assuming its inner matrix compartment is a sphere of diameter 1.5 μm.

(d) From these data, is the pH gradient alone sufficient to generate ATP?

(e) If not, suggest how the necessary energy for synthesis of ATP arises.

Answer

(a) Using the equation $pH = -\log [H^+]$, we can calculate external $[H^+] = 10^{-7.4} = 4.0 \times 10^{-8}$ M; and internal $[H^+] = 10^{-7.7} = 2.0 \times 10^{-8}$ M.

(b) From **(a),** the ratio is 2:1. We can calculate the free energy inherent in this *concentration* difference across the membrane. Assuming a temperature of 25 °C:

$$\begin{aligned}\Delta G &= RT \ln (C_2/C_1)\\ &= (2.48 \text{ kJ/mol}) \ln 2\\ &= -1.7 \text{ kJ/mol}\end{aligned}$$

(c) Given that the volume of the mitochondrion $= \frac{4}{3}\pi(0.75 \times 10^{-3} \text{ mm})^3$ and $[H^+] = 2.0 \times 10^{-8}$ M, the number of protons is

$$\frac{(1.33)(3.14)(0.75 \times 10^{-3} \text{ mm})^3(2.0 \times 10^{-8} \text{ mol/L})(6.02 \times 10^{23} \text{ protons/mol})}{(10^6 \text{ mm}^3\text{/L})} = 21 \text{ protons}$$

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(d) No; the energy available from the H^+ concentration gradient, $2.3\Delta pH\,RT = 2.3(0.3)$ (2.48 kJ/mol) = 1.7 kJ/mol, is insufficient to synthesize 1 mol of ATP.

(e) The total energy inherent in the pH gradient is the sum of the energy due to the concentration gradient and the energy due to the charge separation. The overall transmembrane electrical potential is the main factor in producing a sufficiently large ΔG_t (see Eqns 19–8 and 19–9).

17. Rate of ATP Turnover in Rat Heart Muscle Rat heart muscle operating aerobically fills more than 90% of its ATP needs by oxidative phosphorylation. Each gram of tissue consumes O_2 at the rate of 10.0 μmol/min, with glucose as the fuel source.

(a) Calculate the rate at which the heart muscle consumes glucose and produces ATP.

(b) For a steady-state concentration of ATP of 5.0 μmol/g of heart muscle tissue, calculate the time required (in seconds) to completely turn over the cellular pool of ATP. What does this result indicate about the need for tight regulation of ATP production? (Note: Concentrations are expressed as micromoles per gram of muscle tissue because the tissue is mostly water.)

Answer ATP turns over very rapidly in all types of tissues and cells.

(a) Glucose oxidation requires 6 mol of O_2 per mol of glucose. Therefore, glucose is consumed at the rate of (10.0 μmol/min · g)/6 = 1.7 μmol/min · g of tissue. If each glucose produces 32 ATP (see Table 19–5), the muscle produces ATP at the rate of (1.7 μmol glucose/min · g)(32 ATP/glucose) = 54 μmol/min · g, or 0.91 μmol/s · g.

(b) It takes (5.0 μmol/g)/(0.91 μmol/s · g) = 5.5 s to produce 5.0 μmol of ATP per gram, so the entire pool of ATP must be regenerated (turned over) every 5.5 s. In order to do this, the cell must regulate ATP synthesis precisely.

18. Rate of ATP Breakdown in Flight Muscle ATP production in the flight muscle of the fly *Lucilia sericata* results almost exclusively from oxidative phosphorylation. During flight, 187 mL of O_2/hr · g of body weight is needed to maintain an ATP concentration of 7.0 μmol/g of flight muscle. Assuming that flight muscle makes up 20% of the weight of the fly, calculate the rate at which the flight-muscle ATP pool turns over. How long would the reservoir of ATP last in the absence of oxidative phosphorylation? Assume that reducing equivalents are transferred by the glycerol 3-phosphate shuttle and that O_2 is at 25 °C and 101.3 kPa (1 atm).

Answer Using the gas laws ($PV = nRT$), we can calculate that 187 mL of O_2 contains

$$n = PV/RT = (1\text{ atm})(0.187\text{ L})/(0.08205\text{ L} \cdot \text{atm/mol} \cdot \text{K})(298\text{ K}) = 7650\ \mu\text{mol of } O_2$$

Thus, the rate of oxygen consumption by flight muscle is

$$(7650\ \mu\text{mol/hr})/(1\text{ g})(0.2)(3600\text{ s/hr}) = 10.6\ \mu\text{mol/s} \cdot \text{g}$$

Assuming a yield of 30 ATP per glucose (see Table 19–5; assume the use of the glycerol 3-phosphate shuttle), and given 6 O_2 consumed per glucose, the amount of ATP formed is

$$[(30\text{ ATP/glucose})/(6\ O_2\text{/glucose})](10.6\ \mu\text{mol } O_2\text{/s} \cdot \text{g}) = 53\ \mu\text{mol/s} \cdot \text{g}$$

Thus, a reservoir of 7.0 μmol/g would last (7.0 μmol/g)/(53 μmol/s · g) = 0.13 s.



19. Mitochondrial Disease and Cancer Mutations in the genes that encode certain mitochondrial proteins are associated with a high incidence of some types of cancer. How might defective mitochondria lead to cancer?

Answer Reactive oxygen species react with macromolecules, including DNA. If a mitochondrial defect leads to increased production of ROS, the nuclear genes that encode proto-oncogenes (pp. 473, 474) can be damaged, producing oncogenes and leading to unregulated cell division and cancer.

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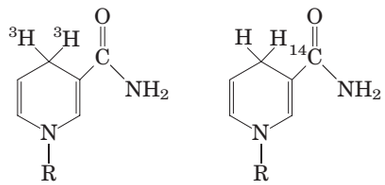
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20. Variable Severity of a Mitochondrial Disease Individuals with a disease caused by a specific defect in the mitochondrial genome may have symptoms ranging from mild to severe. Explain why.

Answer The explanation is probably heteroplasmy. In some individuals, a mitochondrial mutation may affect only a small proportion of cells and tissues, because most mitochondria in these cells and tissues have normal genomes and the few mutant mitochondria do not significantly compromise the ability to produce ATP. In other individuals, the chance distribution of mitochondrial genomes during cell division has led to a high degree of heteroplasmy in which many cells have a majority of defective mitochondria, resulting in more severe symptoms.

21. Transmembrane Movement of Reducing Equivalents Under aerobic conditions, extramitochondrial NADH must be oxidized by the mitochondrial electron-transfer chain. Consider a preparation of rat hepatocytes containing mitochondria and all the cytosolic enzymes. If [4-^{3}H]NADH is introduced, radioactivity soon appears in the mitochondrial matrix. However, if [7-^{14}C]NADH is introduced, no radioactivity appears in the matrix. What do these observations reveal about the oxidation of extramitochondrial NADH by the electron-transfer chain?



[4-^{3}H]NADH [7-^{14}C]NADH

Answer The malate-aspartate shuttle transfers electrons and protons from the cytoplasm into the mitochondrion. Neither NAD^+ nor NADH passes through the inner membrane, thus the labeled NAD moiety of [7-^{14}C]NADH remains in the cytosol. The ^{3}H on [4-^{3}H]NADH enters the mitochondrion via the malate-aspartate shuttle (see Fig. 19–29). In the cytosol, [4-^{3}H]NADH transfers its ^{3}H to oxaloacetate to form [^{3}H]malate, which enters the mitochondrion via the malate–α-ketoglutarate transporter, then donates the ^{3}H to NAD^+ to form [4-^{3}H]NADH in the matrix.

22. High Blood Alanine Level Associated with Defects in Oxidative Phosphorylation Most individuals with genetic defects in oxidative phosphorylation are found to have relatively high concentrations of alanine in their blood. Explain this in biochemical terms.

Answer In these individuals, the usual route for pyruvate metabolism—conversion to acetyl-CoA and entry into the citric acid cycle—is slowed by the decreased capacity for carrying electrons from NADH to oxygen. Accumulation of pyruvate in the tissues shifts the equilibrium for pyruvate-alanine transaminase, resulting in elevated concentrations of alanine in tissues and blood.

23. NAD Pools and Dehydrogenase Activities Although both pyruvate dehydrogenase and glyceraldehyde 3-phosphate dehydrogenase use NAD^+ as their electron acceptor, the two enzymes do not compete for the same cellular NAD pool. Why?

Answer Pyruvate dehydrogenase is located in the mitochondrion, and glyceraldehyde 3-phosphate dehydrogenase in the cytosol. Because the mitochondrial and cytosolic pools of NAD are separated by the inner mitochondrial membrane, the enzymes do not compete for the same NAD pool. However, reducing equivalents are transferred from one nicotinamide coenzyme pool to the other via shuttle mechanisms (see Problem 21).

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24. Diabetes as a Consequence of Mitochondrial Defects Glucokinase is essential in the metabolism of glucose in pancreatic β cells. Humans with two defective copies of the glucokinase gene exhibit a severe, neonatal diabetes, whereas those with only one defective copy of the gene have a much milder form of the disease (mature onset diabetes of the young, MODY2). Explain this difference in terms of the biology of the β cell.

Answer In β cells in which both copies of glucokinase are defective, the rate of glycolytic ATP production does not increase when blood glucose rises, and thus these cells cannot produce high enough concentrations of ATP to affect the ATP-dependent K^+ channel that indirectly regulates insulin secretion. With one functional copy of glucokinase, β cells can respond to very high glucose concentrations by producing suprathreshold concentrations of ATP, triggering insulin release.

25. Effects of Mutations in Mitochondrial Complex II Single nucleotide changes in the gene for succinate dehydrogenase (Complex II) are associated with midgut carcinoid tumors. Suggest a mechanism to explain this observation.

Answer Defects in Complex II result in increased production of ROS, damage to DNA, and mutations that lead to unregulated cell division (cancer). It is not clear why the cancer tends to occur in the midgut.

26. Photochemical Efficiency of Light at Different Wavelengths The rate of photosynthesis, measured by O_2 production, is higher when a green plant is illuminated with light of wavelength 680 nm than with light of 700 nm. However, illumination by a combination of light of 680 nm and 700 nm gives a higher rate of photosynthesis than light of either wavelength alone. Explain.

Answer Plants have two photosystems. Photosystem I absorbs light maximally at 700 nm and catalyzes cyclic photophosphorylation and $NADP^+$ reduction (see Fig. 19–56). Photosystem II absorbs light maximally at 680 nm, splits H_2O to O_2 and H^+, and donates electrons and H^+ to PSI. Therefore, light of 680 nm is better in promoting O_2 production, but maximum photosynthetic rates are observed only when plants are illuminated with light of both wavelengths.

27. Balance Sheet for Photosynthesis In 1804 Theodore de Saussure observed that the total weights of oxygen and dry organic matter produced by plants is greater than the weight of carbon dioxide consumed during photosynthesis. Where does the extra weight come from?

Answer Because the general reaction for plant photosynthesis is

$$CO_2 + H_2O \longrightarrow O_2 + \text{organic matter}$$

the extra weight must come from the water consumed in the overall reaction.

28. Role of H_2S in Some Photosynthetic Bacteria Illuminated purple sulfur bacteria carry out photosynthesis in the presence of H_2O and $^{14}CO_2$, but only if H_2S is added and O_2 is absent. During the course of photosynthesis, measured by formation of [^{14}C]carbohydrate, H_2S is converted to elemental sulfur, but no O_2 is evolved. What is the role of the conversion of H_2S to sulfur? Why is no O_2 evolved?

Answer Purple sulfur bacteria use H_2S as a source of electrons and protons:

$$H_2S \longrightarrow S + 2H^+ + 2e^-$$

The electrons are "activated" by a light energy–capturing photosystem. These cells produce their ATP by photophosphorylation and their NADPH from H_2S oxidation. Because H_2O is not split, O_2 is not evolved (photosystem II is absent).

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29. Boosting the Reducing Power of Photosystem I by Light Absorption When photosystem I absorbs red light at 700 nm, the standard reduction potential of P700 changes from 0.40 V to about −1.2 V. What fraction of the absorbed light is trapped in the form of reducing power?

Answer For a change in standard reduction potential, ΔE, of 0.4 V − (−1.2 V), the free-energy change per electron is

$$\begin{aligned}\Delta G'^\circ &= n\mathcal{F}\Delta E'^\circ \\ &= -(96.48\ \text{kJ/V} \cdot \text{mol})(-1.6\ \text{V}) \\ &= 150\ \text{kJ/mol}\end{aligned}$$

Two photons are absorbed per electron elevated to a higher energy level, which for 700 nm light is equivalent to 2(170 kJ/mol) = 340 kJ/mol (see Fig. 19–46). Thus, the fraction of light energy trapped as reducing power is

$$(150\ \text{kJ/mol})/(340\ \text{kJ/mol}) = 0.44, \text{ or } 44\%$$

30. Electron Flow through Photosystems I and II Predict how an inhibitor of electron passage through pheophytin would affect electron flow through **(a)** photosystem II and **(b)** photosystem I. Explain your reasoning.

Answer

(a) Electron flow through PSII would stop, as there would be no way for electrons to move from PSII to the cytochrome b_6f complex and all the electron acceptors in PSII would very quickly be reduced.

(b) Electron flow through PSI would probably slow. The supply of electrons from PSII would be blocked, but with operation of the cyclic pathway, some electron flow through PSI could continue.

31. Limited ATP Synthesis in the Dark In a laboratory experiment, spinach chloroplasts are illuminated in the absence of ADP and P_i, then the light is turned off and ADP and P_i are added. ATP is synthesized for a short time in the dark. Explain this finding.

Answer Illumination of chloroplasts in the absence of ADP and P_i sets up a proton gradient across the thylakoid membrane. When ADP and P_i are added, ATP synthesis is driven by the gradient. In the absence of continuous illumination, the gradient soon becomes exhausted and ATP synthesis stops.

32. Mode of Action of the Herbicide DCMU When chloroplasts are treated with 3-(3,4-dichlorophenyl)-1,1-dimethylurea (DCMU, or diuron), a potent herbicide, O_2 evolution and photophosphorylation cease. Oxygen evolution, but not photophosphorylation, can be restored by addition of an external electron acceptor, or Hill reagent. How does DCMU act as a weed killer? Suggest a location for the inhibitory action of this herbicide in the scheme shown in Figure 19–56. Explain.

Answer DCMU must inhibit the electron-transfer system linking photosystem II and photosystem I at a position ahead of the first site of ATP production. DCMU competes with PQ_B for electrons from PQ_A (Table 19–4). Addition of a Hill reagent allows H_2O to be split and O_2 to be evolved, but electrons are pulled out of the system before the point of ATP synthesis and before the production of NADPH. DCMU kills plants by inhibiting ATP production.

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33. Effect of Venturicidin on Oxygen Evolution Venturicidin is a powerful inhibitor of the chloroplast ATP synthase, interacting with the CF_o part of the enzyme and blocking proton passage through the CF_oCF_1 complex. How would venturicidin affect oxygen evolution in a suspension of well-illuminated chloroplasts? Would your answer change if the experiment were done in the presence of an uncoupling reagent such as 2,4-dinitrophenol (DNP)? Explain.

Answer Oxygen evolution requires continuing passage of electrons through PSII. Electrons will continue to flow through PSII and the cytochrome b_6f complex until the energetic cost of pumping a proton across the thylakoid membrane exceeds the energy available from absorption of a photon. This point is soon reached when proton flow through CF_oCF_1 is blocked by venturicidin, and oxygen evolution ceases. Addition of an uncoupling agent provides a route for protons to move through the thylakoid membrane, dissipating the energy of the proton gradient. Electrons can now continue to move through PSII and the cytochrome b_6f complex, and oxygen is produced in the water-splitting reaction.

34. Bioenergetics of Photophosphorylation The steady-state concentrations of ATP, ADP, and P_i in isolated spinach chloroplasts under full illumination at pH 7.0 are 120.0, 6.0, and 700.0 μM, respectively.

(a) What is the free-energy requirement for the synthesis of 1 mol of ATP under these conditions?

(b) The energy for ATP synthesis is furnished by light-induced electron transfer in the chloroplasts. What is the minimum voltage drop necessary (during transfer of a pair of electrons) to synthesize ATP under these conditions? (You may need to refer to Eqn 13–7, p. 515.)

Answer

(a) $$\Delta G = \Delta G'^\circ + RT \ln \frac{[\text{ATP}]}{[\text{ADP}][P_i]}$$

$$= 30.5 \text{ kJ/mol} + (2.48 \text{ kJ/mol}) \ln \frac{1.2 \times 10^{-4}}{(6.0 \times 10^{-6})\,(7.0 \times 10^{-4})}$$

$$= 30.5 \text{ kJ/mol} + 25.4 \text{ kJ/mol} = 55.9 \text{ kJ/mol}$$

$$= 56 \text{ kJ/mol (two significant figures)}$$

(b) $$\Delta G = -n\mathcal{F}\Delta E$$

$$\Delta E = -\Delta G/n\mathcal{F}$$

$$= \frac{-56 \text{ kJ/mol}}{-2(96.48 \text{ kJ/V} \cdot \text{mol})}$$

$$= 0.29 \text{ V}$$

35. Light Energy for a Redox Reaction Suppose you have isolated a new photosynthetic microorganism that oxidizes H_2S and passes the electrons to NAD^+. What wavelength of light would provide enough energy for H_2S to reduce NAD^+ under standard conditions? Assume 100% efficiency in the photochemical event, and use E'° of -243 mV for H_2S and -320 mV for NAD^+. See Figure 19–46 for the energy equivalents of wavelengths of light.

Answer First, calculate the standard free-energy change ($\Delta G'^\circ$) of the redox reaction

$$NAD^+ + H_2S \longrightarrow NADH + S^- + H^+$$

Because $\Delta E'^\circ = -320 \text{ mV} - (-243 \text{ mV}) = -77 \text{ mV}$,

$$\Delta G'^\circ = -n\mathcal{F}\Delta E'^\circ$$

$$= (-2)(96.48 \text{ kJ/V} \cdot \text{mol})(-0.077 \text{ V}) = 15 \text{ kJ/mol}$$

This is the minimum energy needed to drive the reduction of NAD^+ by H_2S. Inspection of Figure 19–46 shows that the energy in a "mole" of photons (an einstein) in the visible part of the spectrum ranges from 170 to 300 kJ. Any visible light should have sufficient energy to drive the reduction of NADH by H_2S. In principle, and assuming 100% efficiency, even infrared light should have enough energy to drive this reaction.

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36. Equilibrium Constant for Water-Splitting Reactions The coenzyme $NADP^+$ is the terminal electron acceptor in chloroplasts, according to the reaction

$$2H_2O + 2NADP^+ \longrightarrow 2NADPH + 2H^+ + O_2$$

Use the information in Table 19–2 to calculate the equilibrium constant for this reaction at 25 °C. (The relationship between K'_{eq} and $\Delta G'^\circ$ is discussed on p. 492.) How can the chloroplast overcome this unfavorable equilibrium?

Answer Using standard reduction potentials from Table 19–2, $\Delta E'^\circ$ for the reaction is $-0.324\text{ V} - 0.816\text{ V} = -1.140\text{ V}$.

$$\begin{aligned}\Delta G'^\circ &= -n\mathcal{F}\Delta E'^\circ \\ &= -4(96.48\text{ kJ/V}\cdot\text{mol})(-1.140\text{ V}) \\ &= 440\text{ kJ/mol}\end{aligned}$$

(Note that $n = 4$ because 4 electrons are required to produce 1 mol of O_2.)

$$\begin{aligned}\Delta G'^\circ &= -RT\ln K'_{eq} \\ \ln K'_{eq} &= -\Delta G'^\circ/RT \\ &= (-440\text{ kJ/mol})/(2.48\text{ kJ/mol}) \\ &= -177 \\ K'_{eq} &= e^{-177} = 1.35 \times 10^{-77}\end{aligned}$$

The equilibrium is clearly very unfavorable. In chloroplasts, the input of light energy overcomes this barrier.

37. Energetics of Phototransduction During photosynthesis, eight photons must be absorbed (four by each photosystem) for every O_2 molecule produced:

$$2H_2O + 2NADP^+ + 8\text{ photons} \longrightarrow 2NADPH + 2H^+ + O_2$$

Assuming that these photons have a wavelength of 700 nm (red) and that the absorption and use of light energy are 100% efficient, calculate the free-energy change for the process.

Answer From Problem 36, $\Delta G'^\circ$ for the production of 1 mol of O_2 in this reaction is 440 kJ/mol. A light input of 8 photons (700 nm) is equivalent to (8)(170 kJ/einstein) = 1360 kJ/einstein (see Fig. 19–46). (An einstein is a "mole" of photons.) The overall standard free-energy change of the reaction is

$$\Delta G'^\circ = (440 - 1360)\text{ kJ/mol} = -920\text{ kJ/mol}$$

38. Electron Transfer to a Hill Reagent Isolated spinach chloroplasts evolve O_2 when illuminated in the presence of potassium ferricyanide (a Hill reagent), according to the equation

$$2H_2O + 4Fe^{3+} \longrightarrow O_2 + 4H^+ + 4Fe^{2+}$$

where Fe^{3+} represents ferricyanide and Fe^{2+}, ferrocyanide. Is NADPH produced in this process? Explain.

Answer No NADPH is produced. Artificial electron acceptors can remove electrons from the photosynthetic system and stimulate O_2 production. Ferricyanide competes with the cytochrome b_6f complex for electrons and removes them from the system. Consequently, P700 (of photosystem I) does not receive any electrons that can be activated for $NADP^+$ reduction. However, O_2 is evolved because all components of photosystem II are oxidized (see Fig. 19–56).

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39. How Often Does a Chlorophyll Molecule Absorb a Photon? The amount of chlorophyll *a* (M_r 892) in a spinach leaf is about 20 μg/cm^2 of leaf. In noonday sunlight (average energy reaching the leaf is 5.4 J/cm^2 · min), the leaf absorbs about 50% of the radiation. How often does a single chlorophyll molecule absorb a photon? Given that the average lifetime of an excited chlorophyll molecule in vivo is 1 ns, what fraction of the chlorophyll molecules are excited at any one time?

Answer The leaf absorbs light in units of photons that vary in energy from 170 to 300 kJ/einstein, depending on wavelength (see Fig. 19–46). The leaf absorbs light energy at the rate of $0.5(5.4 \text{ J/cm}^2 \cdot \text{min}) = 2.7 \text{ J/cm}^2 \cdot \text{min}$. Assuming an average energy of 270 kJ/einstein, this rate of light absorption is

$$(2.7 \times 10^{-3} \text{ kJ/cm}^2 \cdot \text{min})/(270 \text{ kJ/einstein}) = 1 \times 10^{-5} \text{ einstein/cm}^2 \cdot \text{min}$$

The concentration of chlorophyll in the leaf is

$$(20 \times 10^{-6} \text{ g/cm}^2)/(892 \text{ g/mol}) = 2 \times 10^{-8} \text{ mol/cm}^2$$

Thus, 1 mol of chlorophyll absorbs 1 einstein of photons every

$$(2 \times 10^{-8} \text{ mol/cm}^2)/(1 \times 10^{-5} \text{ einstein/cm}^2 \cdot \text{min}) = 2 \times 10^{-3} \text{ min} = 0.1 \text{ s}$$

Because excitation lasts about 1 ns = 1×10^{-9} s, the fraction of chlorophylls excited at any one time is $(1 \times 10^{-9} \text{ s})/(0.1 \text{ s}) = 1 \times 10^{-8}$, or one in every 10^8 molecules.

40. Effect of Monochromatic Light on Electron Flow The extent to which an electron carrier is oxidized or reduced during photosynthetic electron transfer can sometimes be observed directly with a spectrophotometer. When chloroplasts are illuminated with 700 nm light, cytochrome *f,* plastocyanin, and plastoquinone are oxidized. When chloroplasts are illuminated with 680 nm light, however, these electron carriers are reduced. Explain.

Answer Light at 700 nm activates electrons in P700 and $NADP^+$ is reduced (see Fig. 19–56). This drains all the electrons from the electron-transfer system between photosystems II and I, because light at 680 nm is not available to replace electrons by activating PSII. When light at 680 nm activates PSII (but not PSI), all the carriers between the two systems become reduced because no electrons are excited in PSI.

41. Function of Cyclic Photophosphorylation When the [NADPH]/[$NADP^+$] ratio in chloroplasts is high, photophosphorylation is predominantly cyclic (see Fig. 19–56). Is O_2 evolved during cyclic photophosphorylation? Is NADPH produced? Explain. What is the main function of cyclic photophosphorylation?

Answer Neither O_2 nor NADPH is produced. At high [NADPH]/[$NADP^+$] ratios, electron transfer from reduced ferredoxin to $NADP^+$ is inhibited and the electrons are diverted into the cytochrome b_6f complex. These electrons return to P700 and ATP is synthesized by photophosphorylation. Because electrons are not lost from P700, none are needed from PSII. Thus, H_2O is not split and O_2 is not produced. In addition, NADPH is not formed because the electrons return to P700. The function of cyclic photophosphorylation is to produce ATP.

Data Analysis Problem

42. Photophosphorylation: Discovery, Rejection, and Rediscovery In the 1930s and 1940s, researchers were beginning to make progress toward understanding the mechanism of photosynthesis. At the time, the role of "energy-rich phosphate bonds" (today, "ATP") in glycolysis and cellular respiration was just becoming known. There were many theories about the mechanism of photosynthesis, especially about the role of light. This problem focuses on what was then called the "primary photochemical process"—that is, on what it is, exactly, that the energy from captured light produces in the photosynthetic cell. Interestingly, one important part of the modern model of photosynthesis was proposed early on, only to be rejected, ignored for several years, then finally revived and accepted.

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In 1944, Emerson, Stauffer, and Umbreit proposed that "the function of light energy in photosynthesis is the formation of 'energy-rich' phosphate bonds" (p. 107). In their model (hereafter, the "Emerson model"), the free energy necessary to drive both CO_2 fixation *and* reduction came from these "energy-rich phosphate bonds" (i.e., ATP), produced as a result of light absorption by a chlorophyll-containing protein.

This model was explicitly rejected by Rabinowitch (1945). After summarizing Emerson and coauthors' findings, Rabinowitch stated: "Until more positive evidence is provided, we are inclined to consider as more convincing a general argument against this hypothesis, which can be derived from energy considerations. Photosynthesis is eminently a problem of energy *accumulation*. What good can be served, then, by converting light quanta (even those of red light, which amount to about 43 kcal per Einstein) into 'phosphate quanta' of only 10 kcal per mole? This appears to be a start in the wrong direction—toward *dissipation* rather than toward accumulation of energy" (Vol. I, p. 228). This argument, along with other evidence, led to the abandonment of the Emerson model until the 1950s, when it was found to be correct—albeit in a modified form.

For each piece of information from Emerson and coauthors' article presented in (**a**) through (**d**) below, answer the following three questions:

1. How does this information support the Emerson model, in which light energy is used directly by chlorophyll *to make ATP*, and the ATP then provides the energy to drive CO_2 fixation and reduction?
2. How would Rabinowitch explain this information, based on his model (and most other models of the day), in which light energy is used directly by chlorophyll *to make reducing compounds*? Rabinowitch wrote: "Theoretically, there is no reason why *all* electronic energy contained in molecules excited by the absorption of light should not be available for oxidation-reduction" (Vol. I, p. 152). In this model, the reducing compounds are then used to fix and reduce CO_2, and the energy for these reactions comes from the large amounts of free energy released by the reduction reactions.
3. How is this information explained by our modern understanding of photosynthesis?

(a) Chlorophyll contains a Mg^{2+} ion, which is known to be an essential cofactor for many enzymes that catalyze phosphorylation and dephosphorylation reactions.

(b) A crude "chlorophyll protein" isolated from photosynthetic cells showed phosphorylating activity.

(c) The phosphorylating activity of the "chlorophyll protein" was inhibited by light.

(d) The levels of several different phosphorylated compounds in photosynthetic cells changed dramatically in response to light exposure. (Emerson and coworkers were not able to identify the specific compounds involved.)

As it turned out, the Emerson and Rabinowitch models were both partly correct and partly incorrect.

(e) Explain how the two models relate to our current model of photosynthesis.

In his rejection of the Emerson model, Rabinowitch went on to say: "The difficulty of the phosphate storage theory appears most clearly when one considers the fact that, in weak light, eight or ten quanta of light are sufficient to reduce one molecule of carbon dioxide. If each quantum should produce one molecule of high-energy phosphate, the accumulated energy would be only 80–100 kcal per Einstein—while photosynthesis requires *at least* 112 kcal per mole, and probably more, because of losses in irreversible partial reactions" (Vol. 1, p. 228).

(f) How does Rabinowitch's value of 8 to 10 photons per molecule of CO_2 reduced compare with the value accepted today? You need to consult Chapter 20 for some of the information required here.

(g) How would you rebut Rabinowitch's argument, based on our current knowledge about photosynthesis?

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Answer

(a) (1) The presence of Mg^{2+} supports the hypothesis that chlorophyll is directly involved in catalysis of the phosphorylation reaction: ADP + $P_i \rightarrow$ ATP. (2) Many enzymes (or other proteins) that contain Mg^{2+} are not phosphorylating enzymes, so the presence of Mg^{2+} in chlorophyll does not prove its role in phosphorylation reactions. (3) The presence of Mg^{2+} is essential to chlorophyll's photochemical properties: light absorption and electron transfer.

(b) (1) Enzymes catalyze reversible reactions, so an isolated enzyme that can, under certain laboratory conditions, catalyze removal of a phosphoryl group could probably, under different conditions (such as in cells), catalyze addition of a phosphoryl group. So it is plausible that chlorophyll could be involved in the phosphorylation of ADP. (2) There are two possible explanations: the chlorophyll protein is a phosphatase only and does not catalyze ADP phosphorylation under cellular conditions, or the crude preparation contains a contaminating phosphatase activity that is unconnected to the photosynthetic reactions. (3) It is likely that the preparation was contaminated with a nonphotosynthetic phosphatase activity.

(c) (1) This light inhibition is what one would expect if the chlorophyll protein catalyzed the reaction ADP + P_i + light $\rightarrow$ ATP. Without light, the reverse reaction, a dephosphorylation, would be favored. In the presence of light, energy is provided and the equilibrium would shift to the right, reducing the phosphatase activity. (2) This inhibition must be an artifact of the isolation or assay methods. (3) It is unlikely that the crude preparation methods in use at the time preserved intact chloroplast membranes, so the inhibition must be an artifact.

(d) (1) In the presence of light, ATP is synthesized and other phosphorylated intermediates are consumed. (2) In the presence of light, glucose is produced and is metabolized by cellular respiration to produce ATP, with changes in the levels of phosphorylated intermediates. (3) In the presence of light, ATP is produced and other phosphorylated intermediates are consumed.

(e) Light energy is used to produce ATP (as in the Emerson model) *and* is used to produce reducing power (as in the Rabinowitch model).

(f) The approximate stoichiometry for photophosphorylation (Chapter 19) is that 8 photons yield 2 NADPH and about 3 ATP. Two NADPH and 3 ATP are required to reduce 1 CO_2 (Chapter 20). Thus, at a minimum, 8 photons are required per CO_2 molecule reduced. This is in good agreement with Rabinowitch's value.

(g) Because the energy of light is used to produce *both* ATP and NADPH, each photon absorbed contributes more than just 1 ATP for photosynthesis. The process of energy extraction from light is more efficient than Rabinowitch supposed, and plenty of energy is available for this process—even with red light.

Reference

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